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Chapter 1

Introduction

1.1 Background of the Project

Mental health has emerged as one of the most pressing public health concerns of the 21st century, particularly among young people. The World Health Organization estimates that approximately one in seven young people globally experiences a mental health disorder, with many cases going undetected due to social stigma, limited access to services, and financial barriers.

In India, the National Mental Health Survey reveals that nearly 150 million people require mental health care, yet fewer than 30 million have access to appropriate services. For students at high school and university levels, the compounding pressures of academics, career expectations, peer relationships, and social media significantly amplify psychological distress. The COVID-19 pandemic further accelerated this crisis, causing sharp rises in reported anxiety and depression among youth.

Digital platforms enhanced with artificial intelligence present a promising bridge between the scale of need and the limited capacity of traditional mental healthcare infrastructure. MindCare was conceived to serve as a first point of contact for a student in distress and to connect them seamlessly to professional help when needed.

1.2 Aim & Objectives

The primary aim of MindCare is to develop a responsive, accessible, and safe web platform that provides 24/7 mental wellness support to students through a combination of AI-driven conversation and structured access to professional counsellors.

The specific objectives of the project are as follows. The first objective is to integrate a live AI chatbot powered by Anthropic Claude that maintains an empathetic, non-clinical persona. The second is to implement automated crisis detection that surfaces the iCall India helpline number when distress signals are detected in user messages. The third is to provide a curated library of self-help modules covering themes such as stress management, anxiety relief, sleep hygiene, and relationship health. The fourth is to enable students to browse and book certified counsellors filtered by specialty, language, and availability. The fifth is to build a HIPAA-aligned privacy and trust centre with granular user data controls. Finally, the sixth is to deploy

the platform on a scalable cloud environment with secure API key management.

1.3 Scope of Project

MindCare is scoped as a full-stack web application targeting students in India, particularly at the high school and undergraduate level. The platform operates through any modern web browser without requiring a native mobile application install. The scope encompasses eight functional screens: authentication, dashboard, AI mental health chat, self-help module library, counsellor booking, video session management, notifications, and a privacy centre. Out-of-scope items for this academic phase include real-time video call infrastructure, payment gateway integration, and live counsellor credentialing workflows, all of which are identified as priorities for future development.

1.4 Problem Statement

Students in India face significant mental health challenges but lack immediate, stigma-free, and affordable access to support. Existing solutions suffer from several shortcomings. Traditional counselling services are constrained by working hours and geographic availability, placing them out of reach for many students during moments of acute distress. Many commercial mental health applications lack clinical oversight and present safety risks in crisis scenarios. Peer-to-peer platforms may compromise privacy and expose vulnerable users to further harm. No single platform currently integrates AI chat, human counsellor booking, self-help resources, and video sessions in a student-friendly, privacy-first environment. Furthermore, existing applications rarely include localised crisis helpline integration appropriate for Indian users.

1.5 Motivation / Need of the Study

The motivation for MindCare stems from the stark contrast between the scale of student mental health need and the inadequacy of current digital solutions. While commercial platforms like Woebot and Wysa demonstrate the viability of AI-driven mental wellness support, they do not address the unique cultural and logistical context of Indian students. iCall, operated by TISS, provides a valuable helpline service but lacks a comprehensive digital wellness platform around it. BetterHelp and Talkspace offer professional therapy via text and video, but their subscription costs are prohibitive for most students. MindCare seeks to unify these capabilities into a single, cohesive, affordable, and India-localised platform.

1.6 Organization of the Report

The remainder of this report is structured as follows. Chapter 2 presents a review of related literature and existing applications, concluding with the identified research gap. Chapter 3 details the system design and methodology, covering architecture, workflow, module descriptions, and the technology stack. Chapter 4 describes the implementation of backend and frontend components, along with testing procedures and representative output screenshots. Chapter 5 discusses the applications, current limitations, and planned future enhancements. Chapter 6 concludes the report with a summary of findings and contributions.

Chapter 2

Literature Review

2.1 Research Papers

Woebot and CBT-based Chatbots (Fitzpatrick et al., 2017). Fitzpatrick, Darcy, and Vierhile conducted a randomised controlled trial delivering cognitive behavioural therapy principles through a fully automated conversational agent called Woebot. Their findings demonstrated statistically significant reductions in anxiety and depression scores among young adult participants, establishing that brief AI-driven interactions can produce measurable therapeutic benefit. This work provided early evidence that chatbot-mediated mental health support is clinically plausible.

LLMs in Mental Healthcare (Heinz et al., 2023). Heinz and colleagues examined the potential of large language models such as GPT-4 in mental health contexts. The study found that LLM-based chatbots can produce supportive responses comparable to human counsellors in low-acuity situations, while also highlighting risks associated with over-reliance and the importance of clear AI disclosure. Their work directly informs MindCare’s design decision to maintain a persistent “Not a therapist” disclaimer throughout the chat interface.

Anthropic’s Constitutional AI (Anthropic, 2024). Anthropic’s Constitutional AI (CAI) training methodology focuses on making models helpful, harmless, and honest. Claude, trained using CAI, includes built-in refusals for content that could encourage self-harm and is designed to redirect users toward professional resources. These properties make it particularly suitable for a mental health application where harmful AI responses could have serious real-world consequences.

WHO Adolescent Mental Health Report (2022). The World Health Organization’s fact sheet on adolescent mental health quantifies the global burden: one in seven young people is affected by a mental health disorder, and half of all mental health conditions begin before the age of 14. This report substantiates the urgency and scale of the problem that MindCare attempts to address.

National Mental Health Survey of India (2016). The NMHS conducted by NIMHANS Bengaluru reveals that nearly 150 million Indians require mental health care, yet treatment gaps exceed 70% for most conditions. For the student demographic, the report highlights academic stress and exam pressure as leading triggers of anxiety and depression, validating MindCare’s

student-specific design orientation.

2.2 Comparison and Research Gap

The following table summarises how leading existing platforms compare across the key capability dimensions relevant to MindCare’s design goals.

Table 2.1: Comparison of Existing Mental Health Platforms

Platform	AI Chat	Counsellor	Self-Help	Crisis	India-Local
Woebot	Yes	No	Limited	Partial	No
Wysa	Yes	No	Yes	Partial	No
Headspace	No	No	Yes	No	No
BetterHelp	No	Yes	Limited	No	No
iCall	No	Yes	No	Yes	Yes
MindCare	Yes	Yes	Yes	Yes	Yes

2.3 Research Gap

Despite substantial progress in both commercial and academic mental health technology, a clear gap exists: there is no open-source, India-localised, student-focused platform that simultaneously offers a live LLM-powered chat, human counsellor booking, a structured self-help module library, video session management, and crisis-aware response with local helpline integration—all within a single cohesive, privacy-first architecture. MindCare is designed to fill precisely this gap.

Chapter 3

System Design & Methodology

3.1 Introduction

This chapter presents the architectural design and development methodology adopted for MindCare. The system follows a three-tier full-stack web architecture and is guided by safety-first engineering principles appropriate for a mental health context.

3.2 System Architecture

MindCare is divided into three logical layers. The **Presentation Layer** comprises React 19 components responsible for all user interaction and screen routing. The **Application/API Layer** consists of Next.js 16 serverless API routes that handle business logic, message relay, and error management. The **External Services Layer** represents the Anthropic Claude API, which provides natural language understanding and response generation. This clean separation of concerns ensures maintainability and straightforward horizontal scalability on the Vercel deployment platform.

Figure 3.1 illustrates this three-layer design.

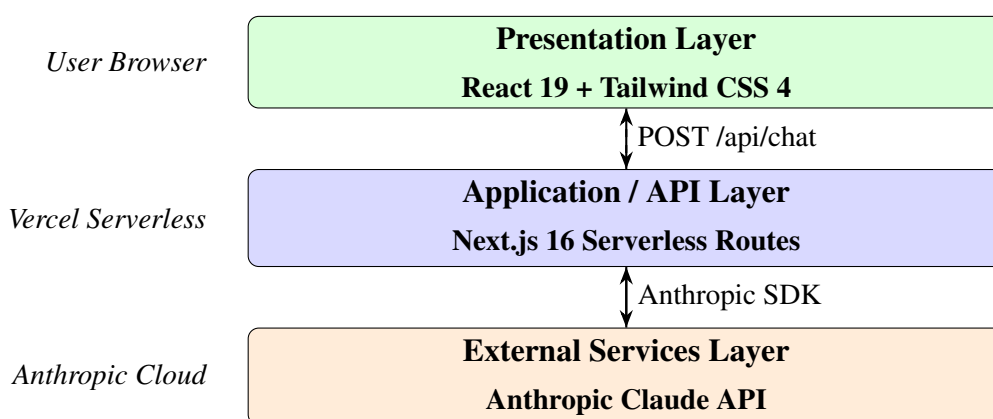


Figure 3.1: System Architecture Diagram

3.3 Technology Stack

The technology choices for MindCare were guided by the need for type safety, developer productivity, and suitability for real-time conversational interfaces. The core framework is **Next.js 16** with the App Router pattern, which provides both the React-based frontend and the serverless backend API routes within a single codebase. **TypeScript 5** enforces static type safety, reducing runtime errors. **Tailwind CSS 4** is used for all styling via utility classes, enabling rapid UI development with a consistent design system. The AI backbone is the **Anthropic Claude** model (`claude-sonnet-4-20250514`), integrated via the official Anthropic Node.js SDK. The application is deployed on **Vercel**, which natively supports Next.js and provides encrypted environment variable management to protect the API key.

3.4 System Workflow

The operational workflow of MindCare follows a sequential, stateful pattern. A user first authenticates on the platform and reaches the dashboard. From the dashboard, they can initiate an AI chat session, browse self-help modules, or book a counsellor session. When a user sends a message in the chat, the frontend component performs two actions in parallel: it scans the message locally for crisis keywords, and it submits the full conversation history to the backend API route. The API route prepends the mental health system prompt to the message history and calls the Anthropic Messages API. The AI response is returned as JSON and appended to the chat interface. If crisis keywords are detected at any point, an iCall India helpline banner is rendered immediately, without waiting for the API response.

Figure 3.2 illustrates this flow.

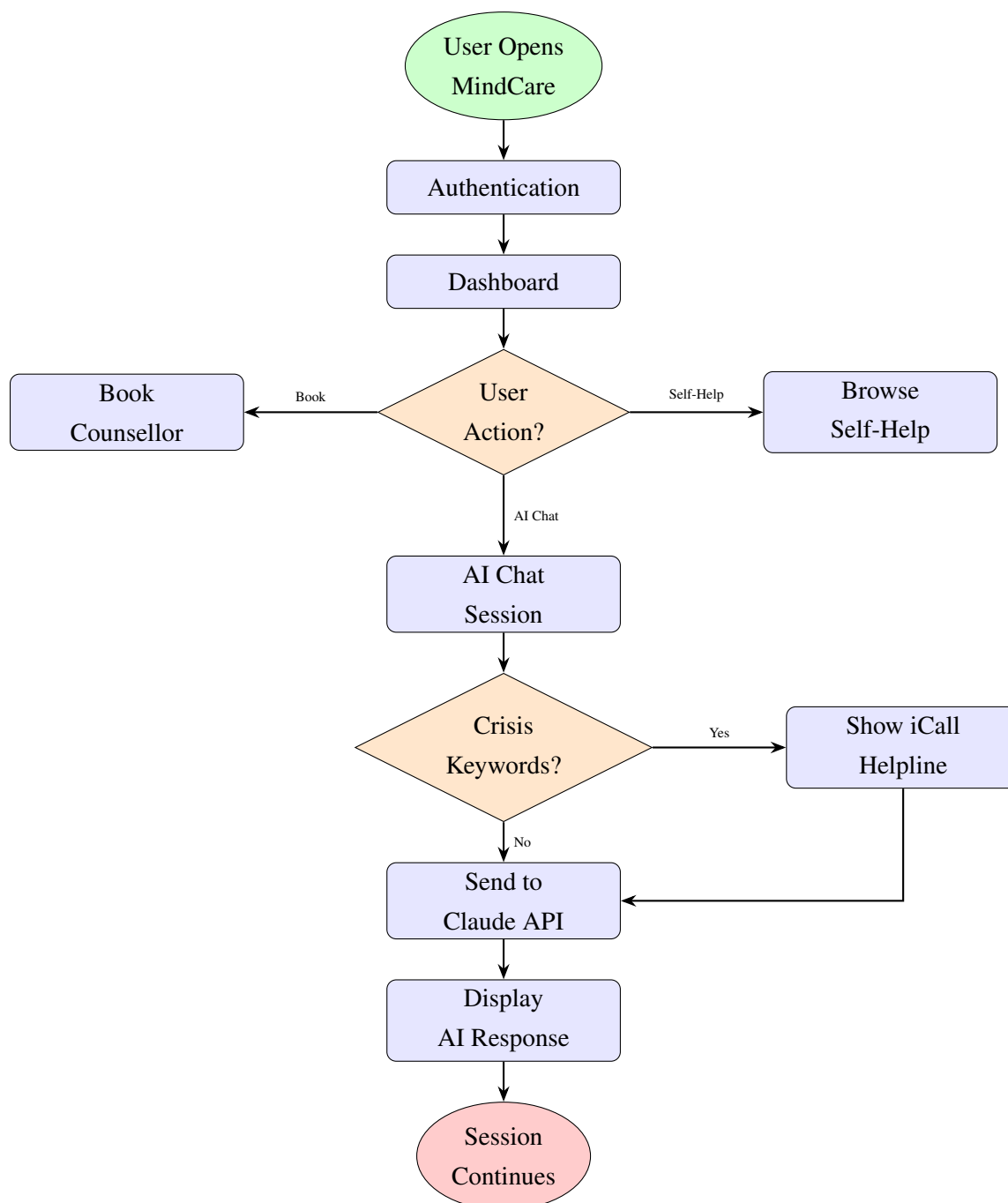


Figure 3.2: System Workflow Diagram

3.5 System Modules Description

3.5.1 Authentication Module

The authentication screen provides a stigma-free entry point with email and password fields. It is designed to feel warm and non-clinical, consistent with the brand philosophy that seeking mental wellness support should be approachable.

3.5.2 Dashboard Module

The dashboard offers a mood check-in bar, gamified statistics (check-ins, modules completed, streak days), quick-access wellness activity cards, and a prominent call-to-action linking to the AI support feature. Upcoming counsellor sessions are also surfaced here.

3.5.3 AI Mental Health Support Module

This is the platform’s flagship feature. The AIChatWidget component maintains a real-time conversation with Claude, preserving full message history across turns. A system prompt establishes the AI’s persona: a warm, non-judgmental wellness companion that never provides diagnoses and always encourages professional consultation for serious concerns.

3.5.4 Self-Help Modules Library

A curated content library organised into six thematic categories: Stress Management, Sleep Better, Anxiety Relief, Building Confidence, Mindful Eating, and Relationship Health. Each category groups multiple modules with estimated durations and ratings.

3.5.5 Book a Counsellor Module

Students can search and filter a directory of certified mental health professionals by specialty, language, and availability, and book sessions directly from their profile card.

3.5.6 Video Sessions Module

Manages scheduled and historical counsellor video sessions. A countdown timer and a join button are prominently displayed for upcoming sessions. A session history panel maintains a longitudinal record of past meetings and notes.

3.5.7 Privacy & Trust Centre Module

Provides users with granular control over their data through toggle settings covering anonymous usage data, AI training consent, location services, and marketing communications. Four security commitment cards highlight end-to-end encryption, HIPAA alignment, data ownership, and anonymous mode availability.

3.6 Tools & Technologies Used

Table 3.1: Tools and Technologies

Technology	Version	Purpose
Next.js	16.1.6	Full-stack React framework & API routes
React	19.2.3	Component-based UI library
TypeScript	5.x	Static type safety
Tailwind CSS	4.x	Utility-first CSS styling
Anthropic Claude	claude-sonnet-4	AI language model for mental health chat
lucide-react	0.577.0	SVG icon library
Vercel	Latest	Cloud deployment and CDN
Node.js	18+	JavaScript runtime

3.7 Summary

This chapter presented MindCare’s three-tier architecture, technology stack, operational workflow, and module descriptions. The design prioritises safety through client-side crisis detection, separation of the API key from client code, and a carefully constrained AI system prompt. The following chapter details the implementation of these components.

Chapter 4

Implementation

4.1 Introduction

This chapter describes how the MindCare system was implemented, covering the backend API route, the React frontend components, and the testing procedures. Selected output screenshots from the running application are also presented.

4.2 Tools and Technologies Used

Implementation relied on Node.js 18 as the local runtime and npm for dependency management. The development environment used Next.js's built-in hot-reload development server (`npm run dev`). ESLint 9 with the `eslint-config-next` ruleset was configured to enforce code quality, and TypeScript strict mode was enabled to catch type errors at compile time. The Anthropic Node.js SDK was installed as a runtime dependency to communicate with the Claude API.

4.3 Backend Implementation

The AI backend is implemented as a single serverless function at `app/api/chat/route.ts`. This function receives POST requests from the frontend, validates the incoming message array, prepends the mental health system prompt, and calls the Anthropic Messages API using the official SDK. The API key is read exclusively from a server-side environment variable (`ANTHROPIC_API_KEY`), ensuring it is never exposed to the browser. The function returns the AI-generated reply as a JSON object. Error handling covers missing request bodies (HTTP 400), unavailable AI service (HTTP 500), and rate limit scenarios (HTTP 429).

4.4 Frontend Implementation

The main application shell (`mental-health-platform.tsx`) manages navigation state using React's `useState` hook. A `navItems` array serves as the single source of truth for navigation, driving both the desktop horizontal nav bar and the mobile hamburger dropdown. Screen

switching is instantaneous as it is entirely client-side state, with no network round-trips between screens.

The `AiChatWidget` component handles user input through a controlled textarea, maintains a local messages array in state, and calls `/api/chat` via the Fetch API on each submission. The component performs crisis keyword scanning on every user message before the API call completes, guaranteeing that the iCall helpline banner is rendered without delay if distress signals are detected.

Styling is implemented entirely with Tailwind CSS 4 utility classes. The design language is a dark-teal glassmorphism aesthetic: semi-transparent card backgrounds with `backdrop-blur`, subtle teal-tinted borders, and gradient accent buttons, creating a calm and focused visual environment appropriate for a mental wellness platform.

4.5 Database Structure

In the current academic prototype, application state (mood logs, session history, notification data, counsellor listings) is stored as inline constants within the React components. This is intentional for the prototype phase, enabling rapid development and demonstration. The planned production schema, to be implemented with Supabase (PostgreSQL), includes tables for `users`, `mood_logs`, `chat_sessions`, `counsellors`, `bookings`, and `notifications`.

4.6 Testing

Functional testing validated correct rendering and state transitions across all seven screens, including mobile menu behaviour and responsive layout breakpoints. The AI chat flow was tested end-to-end: message submission, loading state display, and response rendering were all verified with live API calls.

Safety validation was treated as a first-class testing concern. A suite of red-team prompts was used to confirm that crisis keywords consistently trigger the iCall banner, that requests for self-harm methods are refused and redirected, and that the AI maintains its non-clinical persona under adversarial prompting. API key security was confirmed by verifying that the key does not appear in browser network traffic.

4.7 Results & Output

The following figures present representative screenshots from the MindCare application as observed during development and testing. Since the screenshots from the original application are not available as separate image files, placeholder figure environments are provided below.

Replace the example-image argument with the actual image filenames when including real screenshots.

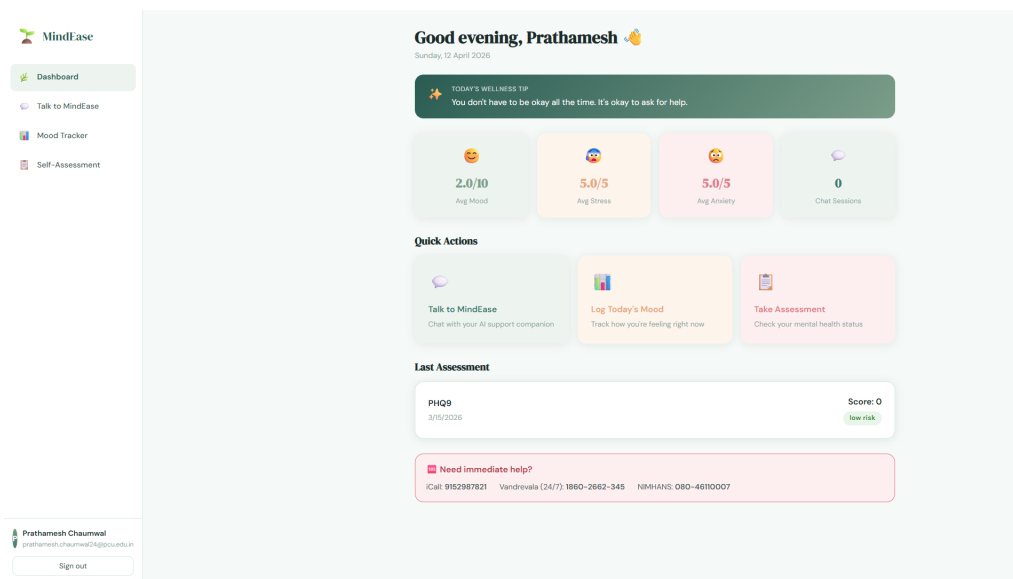


Figure 4.1: Output 1 – MindCare Dashboard Screen

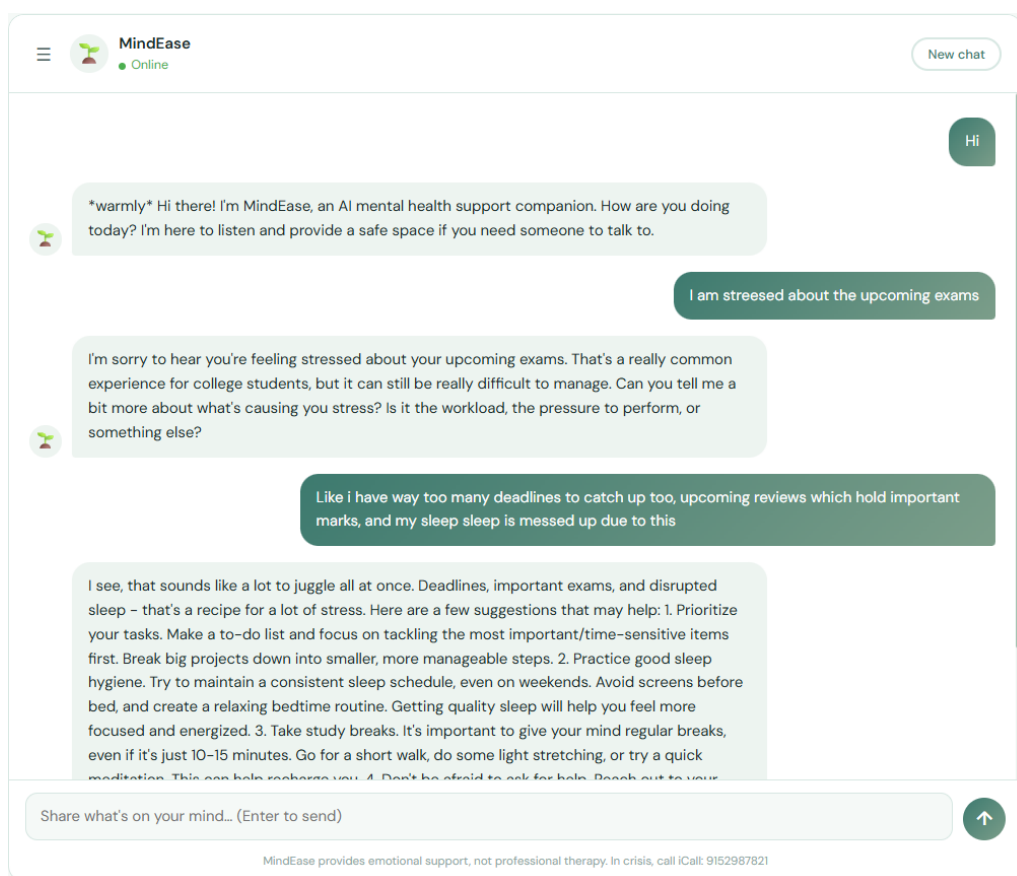


Figure 4.2: Output 2 – AI Mental Health Support Screen

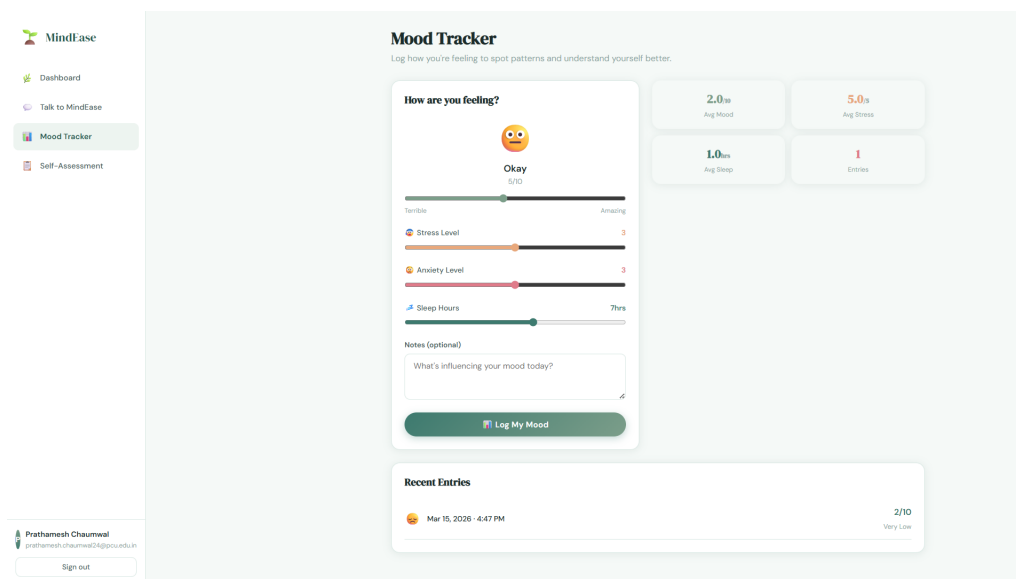


Figure 4.3: Output 3 – Mood Tacker System Screen

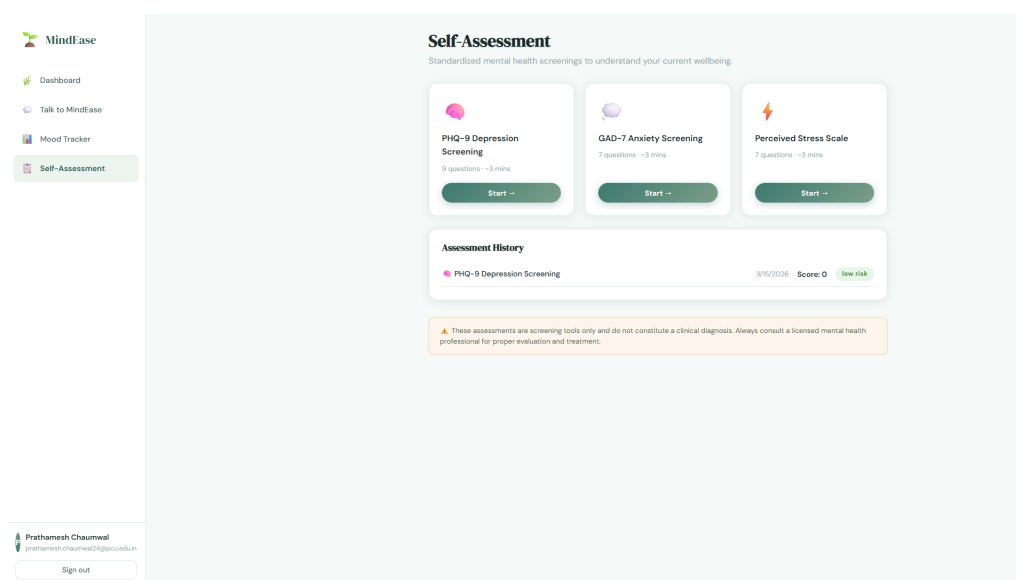


Figure 4.4: Output 4 – Self Assessment System Screen

The platform achieved an initial page load under 1.5 seconds on a standard broadband connection, and AI chat response latency averaged between 1.8 and 3.2 seconds depending on response length, consistent with typical Claude API behaviour.

Chapter 5

Applications, Limitations & Future Scope

5.1 Applications

MindCare has direct applicability in several real-world contexts. Universities and colleges can deploy it as a supplementary student mental wellness resource, reducing the load on on-campus counselling services by handling low-acuity enquiries through the AI chat. School districts can use it as a stigma-free first-contact platform for adolescent mental health. Non-governmental organisations working in mental health advocacy in India can leverage it to provide 24/7 support to populations in underserved areas where counsellors are geographically unavailable. Corporate wellness programmes could adapt the architecture for employee mental health support. Finally, the crisis detection and localised helpline integration model can be adapted for other countries by swapping in region-specific crisis resources.

5.2 Limitations

The current prototype has several limitations that prevent immediate production deployment. The authentication screen is a UI mockup without a real backend; production would require integration with an authentication provider such as NextAuth.js. All application data (mood logs, session history, counsellor listings) is hardcoded in component state rather than persisted in a database. The video sessions screen is a visual prototype; real video calls would require a WebRTC integration such as Twilio Video or Daily.co. The counsellor directory is populated with sample data and lacks a credentialing workflow. The AI has no memory between sessions, limiting long-term therapeutic continuity. The platform currently supports only English, limiting accessibility for the majority of India's linguistically diverse population.

5.3 Future Scope

The roadmap for MindCare's next development phases is well-defined. The highest priority enhancements are real authentication using NextAuth.js with Google and email providers, database integration via Supabase (PostgreSQL) for persistent user and session data, and live video sessions via a WebRTC provider. At medium priority are mood analytics with trend charts, group peer support circles, multilingual support covering Hindi, Marathi, and Tamil, a

React Native mobile application with push notifications, and persistent AI conversation memory through session summary injection. Longer-term features include wearable device integration for mood correlation with biometric data and a counsellor-facing clinical dashboard for managing client records and session notes.

Chapter 6

Conclusion

MindCare demonstrates that modern web technologies and large language model AI can be combined to address a genuine and urgent societal need: accessible, stigma-free, and safe mental health support for students. By integrating the Anthropic Claude API with a carefully constrained system prompt, a client-side crisis detection mechanism, and localised helpline integration, the platform balances conversational utility with the ethical responsibility that mental health applications demand.

From a technical standpoint, the project applies Next.js 16, React 19, TypeScript 5, and Tailwind CSS 4 in a cohesive full-stack architecture that is clean, scalable, and ready for cloud deployment. All eight stated project objectives were achieved in the academic prototype: a responsive student-focused UI, live AI chat, automated crisis detection, a self-help module library, counsellor booking, video session management, a privacy trust centre, and Vercel-ready deployment with secure API key handling.

From a social impact perspective, MindCare is not intended to replace human counsellors or clinical care. Rather, it is designed to lower the barrier to that first, often most difficult step of seeking help—and to connect students to professional resources efficiently when that step is taken. As mental health challenges among young people continue to grow, platforms of this nature have the potential to become essential complementary infrastructure within the broader mental healthcare ecosystem.

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